

have tougher meat due to altered muscle metabolism [20]. To address these challenges, technological solutions are crucial. Providing shade, proper ventilation, and cooling systems can mitigate heat stress while formulating diets with heat-resistant nutrients like electrolytes, antioxidants, and amino acids helps combat heat stress. Feed additives such as probiotics, prebiotics, and enzymes enhance nutrient utilization and gut health. Precision feeding technologies ensure animals receive optimal nutrients based on their specific needs, while genetic selection for heat tolerance and resilience is essential [21]. Adopting sustainable practices, leveraging technology, and prioritizing animal welfare are key strategies to mitigate the adverse effects of climate change on animal agriculture. By implementing these measures, we can ensure resilient livestock and poultry production systems while promoting environmental sustainability [20].

Due to the increasing demand placed on land resources by population growth and the needs of various sectors, the availability of livestock grazing lands is diminishing [22]. Feed availability and the costs for production are essential to the livestock industry in terms of profit and overall sustainability. An estimate of feeding showed a contribution of 60–80% of the variable costs of animal products, such as pork, milk, and poultry production [23–25]. Cost-effective feed rations are, therefore, essential for building resilience in the livestock industry. Feed costs can be managed effectively with knowledge of the vital nutrient requirements for every growth and reproduction stage of a livestock species and the related costs. According to Minson et al. [26] and Underwood [27], relaxation of some nutritional requirements that may not be vital for a particular growth stage can lead to significant reductions in feed costs while maintaining the animal's performance. This was demonstrated by Uyeh et al. [28], using an interactive feed formulation approach based on an evolutionary algorithm.

Animal feed formulation has been defined as allocating feed ingredients that satisfy the minimum requirements of nutrients to achieve specific objectives or achieve the best animal performance regarding yield and weight gain [29,30]. This implies the main aim of feed production is to achieve the optimal composition of ingredients, which results in the desired outcomes when fed to the animal. On the other hand, feed materials are the carriers of nutrients, and feed formulation allows the measurement and combination of several ingredients needed to supply the nutrient requirements of an animal with the desired aim [31]. Also, feed formulation consists of multiple objectives, such as maximizing land use and minimizing feed cost while meeting the appropriate nutrient requirements [28].

Feed formulation techniques have gone through many evolutions. In the 1950s, time-consuming electromechanical machines were used. The 1980s and 1990s brought in the computerized linear programming method to maximize income while considering feed costs. By the turn of the next decade, non-linear programming models were introduced in feed formulation to optimize weight gain in animals and increase yield in products such as milk. Standard practices in the formulation of feed in animal diets are constant formulation, least-cost formulation, and fixed formulation [32].

Constant formulations focus on delivering a consistent amount of nutrient levels. They account for differences in the biological natural feed components and aim to minimize variations in nutrients. The formulation is fixed, involving the addition of a fixed level of feed ingredients and nutrients. This ensures the feed's nutritional content remains constant, regardless of variations in the natural ingredients used. Least-cost formulations, on the other hand, prioritize cost-effectiveness. They involve substituting a higher-cost ingredient with a lower-cost one while still meeting the nutritional requirements of the feed. The goal is to obtain the least-cost formula, hence the name. This method is used for commercial purposes where cost efficiency is a significant factor. While both methods aim to provide nutritionally balanced feed, constant formulations prioritize consistency in nutrient levels, while least-cost formulations prioritize cost efficiency. The choice between these methods depends on the specific goals and constraints of the feed production operation.

Feed formulation comprises two broad techniques: the manual approach and mathematical linear, single/multiple programming methods [31]. The manual approach consists of the Pearson's square method, trial and error method, simultaneous algebraic equations, and the two-by-two matrix technique. Pearson's square method is used to calculate the percentage of two feed ingredients needed to meet the specific nutrient (crude protein, energy, crude fat) level in a mixture. For example, grains such as corn and soybean meal can be used to achieve the crude protein requirement of an animal. It is easy to use; however, the limitation of this technique is that it cannot be used for complex feed mixtures [33,34]. Simultaneous algebraic equations, on the other hand, can simulate two or more feed ingredients to obtain the best possible nutritional content [34]. With this method, the requirements for both protein and energy can be obtained in one mixture, unlike Pearson's square method, which allows for only one nutrient requirement. The trial-and-error method is more prevalent in poultry, and the limitation is that it is tedious and time-consuming [34]. With the two-by-two matrix technique, two feed ingredients are used to solve a two-nutrient requirement problem, the two-by-two matrix is fixed, and a series of equations is calculated to find the solution to the problem [4].

The mathematical linear, single, or multiple programming methods include the single-linear programming approach, a frequent animal diet formulation technique [35]. Waugh [36] made the first attempt at the single-objective programming method to solve a feed-mix problem by optimizing the ration for livestock with a linear program. In linear programming, the model assumes the nutrient requirement as fixed; a few disadvantages stated by Lara and Romeo [37] include that the model's constraints are rigid, can tackle only linear constraints, and can only operate on the objective function. They are also an efficient tool in the decision-making process by solving a series of linear and non-linear programs via interaction with decision-makers. Zhang and Roush [38] mentioned many advantages of using the multiple-objective programming method, including trade-offs in the decision-making process, concurrent action, and tractability. The linear programming method has been the most used mathematical method in the past decades. Its weakness is based on its reliance on cost as the only basis for the decision-maker in feed formulation. These weaknesses can be remedied by decision-making techniques such as goal and multi-objective programming.

2.2. Decision Support Tools in Livestock Feed Formulation

Decision support tools (DSTs) play a crucial role in Livestock Feed Formulation (LFF) by addressing challenges related to diverse data sets and the complexity of monitoring animal-related systems [39]. These tools are essential for achieving two main objectives: improving animal welfare and mitigating greenhouse gas (GHG) emissions. Enhanced animal welfare directly impacts outcomes in livestock production, such as meat, milk, and eggs, as healthy animals yield higher quality and quantity of products. Additionally, optimizing animal welfare contributes to reducing emissions intensity in the production of livestock products, benefiting both farmers' incomes and environmental sustainability [15,40]. Nutrition is a fundamental aspect of animal welfare, evaluated through the Five Domains Model, which emphasizes the importance of providing animals with appropriate feed and water [36,41]. Proper nutrition directly impacts animal health and well-being, influencing growth, reproduction, and immune function. Feed formulation plays a crucial role in determining the nutritional content of animal products, contributing to high-quality meat, milk, and eggs. Additionally, sustainable feed formulation practices promote natural resilience in animals, reduce the need for pharmacological treatments, and minimize environmental impact by reducing nutrient runoff. Prioritizing animal nutrition supports their welfare and ensures the production of safe, high-quality food for consumers, fostering a sustainable agricultural system [41].

Some existing designed decision support tools include diet checks. This method creates a way to estimate whether dairy cows are consuming enough crude protein, metabolized energy, and neutral detergent fiber for a stipulated milk production quantity [42].

Milgen et al. [43] designed and developed another model, the InraPorc tool, to be used by animal nutritionists to evaluate the utilization of nutrients and experiment with different nutritional strategies. Dobos et al. [44] also introduced the FeedPlan decision support tool, which combines parts of managing herd requirements and feed supply as a simple approach to feed planning and budgeting.

Decision support tools are developed to enable the appropriate users to make effective decisions via a clear description of precise steps to undertake and display the various results obtained from the several options executed [45,46]. These tools can be dynamic and non-dynamic software tools with different recommendations specific to the user's inputs. The software tools can help achieve effective farm administration via efficient data recording, analysis, and production of a series of results based on the evidence gathered [47]. Decision support tools in feed formulation can be utilized by applying multiple criteria decision-making techniques. This approach has more advantages over the linear programming technique, such as simultaneously manipulating various conflicting objectives [48].

Rose, D.C. et al. [49] recommend taking into consideration the following factors in designing effective decision support and delivery tools: cost-effectiveness, habit, trust, ease of use, performance expectancy, peer recommendation, relevance to the user, farmer–adviser compatibility, facilitating conditions, information technology, age, the scale of farming, farming type, compliance, and levels of marketing, influencing the use of these tools by the farmers and advisers. The authors concluded that these factors mentioned above could serve as a checklist for a superior design of decision support tools in the future.

2.3. Techniques in Feed Formulation

Trends in feed production, such as advanced models for precision feed production, have the potential to boost livestock productivity and aid in dissipating the pressures it faces. Black, J.R. and Hlubik, J. [50] illustrated the basics of ration formulation via computerized linear programming, outlining the application, upper and lower bounds on feedstuffs, model ratio, and more. Using the linear programming method, a shrimp diet was formulated in Barbieri, M., Cuzon, G. [40]. Eight feed ingredients and eighteen constraints were used in diet formulation, and an optimal solution with a cost reduction of about 30% and no significant loss in the growth performance was achieved. More advanced computer techniques for feed production were developed in the '90s. Lancaster, L.M. [51] developed software for mainframe computers as the foundation for computer-assisted menu planning. Another example is the multi-agent modeling approach Hamel et al. [42] applied to simulate vegetal raw material for poultry production chains. This model represents the multi-faceted workings of decision-making and human input in the feed formulation process. It operates in Java programming language, simulates the processes in production, and delivers the results. Table 1 shows the evolution of feed formulation techniques with the applied models from 1980 to 2020.

Table 1. Feed formulation technique and various models used from 1980 to 2020.

Year	Livestock	Application/Objective	Reference
1980	Beef	A beef cattle ration formulation technique that satisfies the nutrient standards.	[52]
	Lactating cow/growing cattle	Existence of variability in the nutrient requirements of individual animals and errors in feed intake prediction.	[53]
	Mature dairy cows	The integration of current biological knowledge into mathematical models that can be used both as a framework for defining research priorities and for making decisions on dairy cattle feeding.	[54]
	Unspecified	The introduction of multiple-criteria decision-making techniques to agricultural systems modelers and the demonstration of their use in	[48]